Joint Resource Allocation and Trajectory Design for Energy-Efficient UAV Assisted Networks With User Fairness Guarantee

Jing Wang, *Student Member, IEEE*, Haixia Zhang, *Senior Member, IEEE*, Xiaotian Zhou, *Member, IEEE*, Wenjie Liu, *Graduate Student Member, IEEE*, and Dongfeng Yuan, *Senior Member, IEEE*

***Abstract*—This work explores the utilization of rotary-wing unmanned aerial vehicles (UAVs) as aerial base station to provide downlink data services to ground users. The objective is to maximize the energy efficiency (EE) of the UAV assisted communication system while guaranteeing fairness among users. In pursuit of this goal, taking the propulsion power consumption of UAV, the flight constraints of UAV, and the limited communication resources into account, we jointly optimize the power allocation, bandwidth allocation, and trajectory design. Since the corresponding formulated problem is nonconvex, to solve it efficiently, we first addressed its nonsmoothness and then decompose the problem into two subproblems: 1) the joint power and bandwidth assignment and 2) the trajectory design subproblems. We prove that the power allocation and bandwidth assignment subproblem is quasi-convex fractional optimization. A low-complexity iterative algorithm based on Dinkelbach's algorithm and the Lagrange duality is proposed to get optimal solution. Successive convex approximation (SCA) together with Dinkelbach's algorithm are adopted to convert the nonconvex trajectory optimization problem into convex to get a suboptimal solution for UAV trajectory. Through solving the two subproblems iteratively by adopting block coordinate descent (BCD) until convergence, the original problem is solved. Extensive simulation results demonstrate that the proposed algorithm not only has robust convergence properties but also achieves significant improvements in system EE while ensuring user fairness.**

***Index Terms*—Energy efficiency (EE), resource allocation, trajectory optimization, unmanned aerial vehicle (UAV) assisted communications, user fairness.**

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Jing Wang is with the Shandong Key Laboratory of Wireless Communication Technologies, Shandong University, Jinan 250061, Shandong, China, and also with the School of Information Science and Engineering, Shandong University, Qingdao 266237, Shandong, China (e-mail: jing.wang@mail.sdu.edu.cn).

Haixia Zhang, Xiaotian Zhou, and Wenjie Liu are with the Shandong Key Laboratory of Wireless Communication Technologies and the School of Control Science and Engineering, Shandong University, Jinan 250061, Shandong, China (e-mail: haixia.zhang@sdu.edu.cn; xtzhou@sdu.edu.cn; wenjie_liu@mail.sdu.edu.cn).

Dongfeng Yuan is with the Shandong Key Laboratory of Wireless Communication Technologies, Shandong University, Jinan 250061, Shandong, China (e-mail: dfyuan@sdu.edu.cn).

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I. Introduction

UNMANNED aerial vehicles (UAVs) have been deemed as one of the means enhancing communications in the fifth-generation (5G) and beyond 5G systems [1], [2]. Different from terrestrial infrastructure which is statically deployed, the UAVs offer several advantages, such as easier provision of Line-of-Sight (LoS) links, easy deployment, lower cost, and flexible mobile controllability [3], [4], [5]. All these allow UAVs to fit special situations, such as surging traffic demands of the hotspots, e.g., holiday attractions and mall events, as well as emergency situations, e.g., congested roads and disaster scenes [6], [7], [8]. UAVs can serve as aerial base stations (BSs), mobile relays, or charging devices, which also confirms the potential of UAV assisted wireless communications [9].

Despite those multitude of benefits of UAVs, there remains one significant technical challenge in UAV assisted wireless communication. This challenge stems from the limited onboard battery capacity of UAVs, which is restricted by their size and weight, consequently severely restricting the available energy for UAV operations [10]. Therefore, the pursuit of energy-efficient UAV assisted communications has sparked considerable research interest in [11]. To achieve energy-efficient UAV assisted communication, research efforts have been divided into three main categories. The first focuses on reducing the energy consumption of UAV [12], [13], [14], [15], [16], [17], [18]. The second category of work aims to maximize the system utility under the constraints of maximum energy of UAVs [19], [20], [21]. The last body of research tries to maximize the energy efficiency (EE) of the UAV-assisted communications.

To maximize the EE, several studies have focused on the single UAV scenario where the UAV is statically deployed. For example, Chen et al. [22] addressed the joint power control and channel allocation to optimize the overall EE of device-to-device (D2D) pairs, and ensure the secrecy rates for all users. Hu et al. [23] focused on the resource block assignment, transmit power optimization of UAV and D2D transmitters in energy harvesting-powered UAV-assisted D2D networks. Both of these works assume that the UAV hovers statically over the cell. Xie et al. [24] investigated a wireless powered UAV communication system and focused on maximizing EE through jointly optimizing the UAV's position, transmit power, and transmission time of each user device. Sohail et al. [25]

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developed a power-domain nonorthogonal multiple access (NOMA) scheme for a multiservice demands scenario where the joint optimization of transmit power and UAV altitude was studied. None of the above work considers the mobility of UAV, thus the advantage of the flexible mobile controllability of UAV is not utilized.

There are some works have taken into account the mobility of UAV when maximizing the EE. Yang et al. [26] explored a system where one UAV collects data from multiple terrestrial backscattering tags. They assumed that the power consumption during the flight was constant. Li et al. [27], [28] aimed to maximize the EE of UAV-based NOMA networks. The power consumption encompassed transmit power and fixed circuit consumption of the UAV. Li et al. [29] designed a joint optimization framework involving user scheduling, 3-D beamforming, UAV trajectory and power allocation for an emergency wireless communication. We find that when the above work concerns the mobility of the UAV, the propulsion power consumption of the UAV is not considered or reasonably modeled. Since the power consumption of UAV includes both communication power and propulsion power where the latter occupies a large proportion that cannot be ignored.

To address this, some work has investigated the EE maximization in fixed-wing UAV-assisted wireless communication. Zeng and Zhang optimized the UAV trajectory for single UAV assisted single ground user communication [30]. A theoretical model was developed to analyze the propulsion power consumption of the fixed-wing UAV. Ahmed et al. [31] employed a joint optimization framework encompassing the transmission power, trajectory of UAV, UAV-to-user scheduling and assumed the UAV employing a circular trajectory. Ahmed et al. [32] furthered jointly optimized UAV trajectory, transmit power of UAV and BS to realize the energy-efficient UAV relaying communication where an information causality constraint was involved. Li et al. [33] investigated maximizing the EE of the UAV assisted mobile edge computing system through the joint optimization of the computation load allocation, the transmit power of users, and the UAV trajectory. We note that fixed-wing UAVs and rotary-wing UAVs which are the two categories of UAVs concerned in existing work exhibit different propulsion power expressions due to variations in their models, sizes, etc.

Notably, a key distinction for rotary-wing UAVs compared to fixed-wing UAVs is the superior hovering capability, which can provide more degrees of freedom for trajectory design. Thus, some work has focused on maximizing the system's total EE in rotary-wing UAV-assisted communications. For example, Yang et al. [34] investigated an UAV-assisted backscatter communication network to maximize the system's EE. Pan et al. [35], [36] established a multiobjective optimization problem, including simultaneously maximizing service performance of the UAV and minimizing the UAV power consumption to achieve the energy-efficient communication. Duo et al. [3] focused on the secrecy communication, where a UAV collects confidential information from a transmit source while simultaneously sending jamming signals to ground eavesdroppers. Zeng et al. proposed a joint optimization scheme involving user communication scheduling, bandwidth assignment, transmission power, UAV trajectory. Their focus was on maximizing EE in a system with randomly and unevenly distributed user requirements [37]. Cai et al. [38] studied maximizing EE in a novel system where an information UAV served multiple legitimate users while a multiantenna jammer UAV countered multiple ground eavesdroppers. Xu et al. [39] applied the UAV to charge the ground D2D transceivers. They consider maximize the system EE which is the ratio of the total data rate of all D2D users and their circuit and transmit power consumption.

It is important to highlight that when exploring energy-efficient rotary-wing UAV-assisted communications, user fairness is also an important metric to avoid winner-takes-all scenario [40]. Consequently, there are also some works that focus on enhancing system EE while concerning user fairness guarantees. For instance, Lu et al. [40] investigated a downlink UAV-enabled simultaneous wireless information and power transfer system, regarding the minimum count of bits received by users as the criterion for successful transmission. Dai et al. [41] assumed one UAV serving multiple users using time-division multiple access technology and the user scheduling and UAV trajectory are jointly optimized. Wang et al. [42] addressed an EE max-min problem in a space-air-ground downlink relay system, where the joint optimization of UAV trajectory, NOMA scheduling, NOMA power allocation was performed. Masaracchia et al. [43] studied the optimization of power and time allocation over downlink time sharing NOMA communication for maximizing the EE and the fairness among users. Only the UAV transmission power consumption as well as the UAV fixed circuit power consumption are taken into account.

In energy-efficient UAV-assisted ground communications, it is of great importance and necessity to ensure fairness in user services. This means that all users have access to similar communication rates and quality, rather than just a few users occupying most of the resources and other get almost no services. Although there have been some works conducted on maximizing the system EE while ensuring the user service fairness, they either do not take into account propulsion power (it accounts for a large portion of UAV power consumption and has a significant impact on UAV trajectory design), or do not take full advantage of the optimization of multidimensional communication resources as well as the flexible mobility of UAVs. Specifically, the joint optimization of power allocation, bandwidth allocation, and trajectory design to maximize EE while ensuring user fairness in downlink rotary-wing UAV system remains unexplored. Motivated by the above issues, this article investigates a UAV assisted downlink communication system where a single rotary-wing UAV serves multiple ground users. The overall power consumption of system is composed of the propulsion power and communication power of rotary-wing UAV. Considering the limited on-board battery capacity of the UAV, we aim to maximize the EE of the overall system while ensuring the fairness among users to achieve the energy-efficient UAV-assisted communication. Based on the limited bandwidth, limited power resource and limitations on the movement of UAV, we jointly optimize the power allocation, bandwidth allocation and trajectory design. To

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solve the nonconvex optimization problem, we decompose it into two subproblems, joint power and bandwidth allocation and trajectory design. We apply block coordinate descent (BCD) method to optimize these two subproblems iteratively. The simulation results demonstrate favorable convergence of the proposed algorithm. Also, the proposed algorithm has significant system performance advantages compared with other benchmark schemes. The main contributions are summarized as follows.

1) We maximize the EE guaranteeing the user fairness for the rotary-wing UAV-assisted downlink communication networks. To achieve this goal, the objective function is formulated as maximizing the minimum ratio of achievable rate of each user to the overall power consumption of the UAV. The impact of user fairness needs, UAV propulsion power consumption, and communication resource constraints on system performance are all taken into consideration.
2) The power allocation, bandwidth allocation and trajectory optimization is jointly designed to enable energy-efficient communications. We introduce the auxiliary variable to address the nonsmoothness of the joint optimization problem. We solve the formulated problem through decomposing them into two subproblems, the joint power and bandwidth assignment subproblem and the trajectory design subproblem.
3) We prove that the joint power and bandwidth allocation subproblem is a quasi-convex fractional optimization one. To solve it, we propose a low-complexity iterative algorithm by making use of Dinkelbach's algorithm to transform it into a tractable subtraction form and then we apply the Lagrange duality technology to get the optimal solution. The successive convex approximation (SCA) method and Dinkelbach's algorithm are applied to transform the UAV trajectory design subproblem into a convex optimization problem and a suboptimal solution is obtained.
4) We utilize the BCD framework to iteratively solve these two subproblems until convergence to complete the overall algorithmic process. Extensive simulation results show that the proposed algorithm exhibits favorable convergence and can significantly improve the system EE while realizing the best user fairness.

The remainder of this article is structured as follows. The system model and problem formulation are introduced in Section II. In Section III, an efficient iterative algorithm is proposed, which combines block coordinate descent, Lagrangian duality, and SCA techniques to optimize the joint resource allocation and trajectory design. The simulation results, shown in Section IV, evaluate the performance of the designed algorithm. Section V concludes this article.

II. System Model and Problem Formulation

A. Network Model

As illustrated in Fig. 1, we consider a UAV assisted orthogonal frequency-division multiple access (OFDMA) networks where one rotary-wing UAV is leveraged to serve K ground users who want to be served fairly within a circular cell of radius R through the downlink. We assume that all the users are expected fair services, i.e., get similar transmission rate from the UAV. The user set is denoted as $\mathcal{K}$ and $\mathcal{K} \triangleq \{1, \ldots, K\}$. The flight period of the UAV is denoted as T. OFDMA can help UAV to communicate with multiple ground users at the same time without interference, and the orthogonal nature of its subchannels make it easy to allocate bandwidth and transmit power resources to users [44]. We assume there is a central controller mounted on UAV which is capable of collecting user-related information, for example, the locations. The horizontal location of each user k is assumed to be fixed at $\mathbf{w}_k = [x_k, y_k]^T \in \mathbb{R}^{2\times 1}$, $k \in \mathcal{K}$. Meanwhile, the time-varying horizontal location of the UAV at time instant t is represented as $\mathbf{q}_u(t) = [x(t), y(t)]^T \in \mathbb{R}^{2\times 1}$, $0 \le t \le T$. To ensure periodic servicing of users, it is required that the UAV returns to its start position at the end of flight period, i.e., $\mathbf{q}_u(0) = \mathbf{q}_u(T)$. The start position of the UAV for takeoff, denoted as $\mathbf{q}_u(0) = [x_0, y_0]^T$, is predefined. In addition, practical considerations impose a maximum flight speed constraint $V_{\max}$ on the UAV, i.e., $||\dot{\mathbf{q}}_u(t)|| \le V_{\max}$.

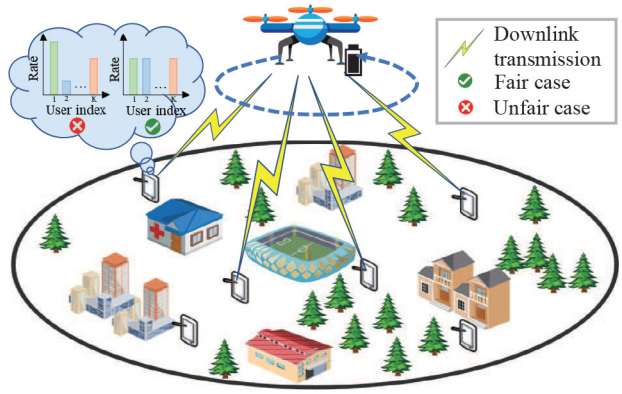

Fig. 1. UAV assisted wireless communication network.

We assume that the UAV maintains a fixed altitude H. To facilitate algorithm design and trajectory planning, we divide the period T into N equally spaced time slots using a discrete trajectory approximation approach. Each time slot has a length of $\phi_t = (T/N)$, which is set to be sufficiently small to ensure that the UAV's location change within each time slot can be negligible. Consequently, the UAV's trajectory over the entire service period is composed of $\mathbf{q}_u[n] = [x(n), y(n)]^T$, where $n = 0, 1, \ldots, N$. It should be noted that $\mathbf{q}_u[0]$ corresponds to the predefined parameter indicating the UAV's starting location. The variables to be optimized are $\mathbf{q}_u[n]$, $n \in \mathcal{N} = 1, \ldots, N$. Thus, the aforementioned constraints can be equivalently expressed as

$$\mathbf{q}_u[0] = \mathbf{q}_u[N] \tag{1}$$

$$\|\mathbf{q}_u[n-1] - \mathbf{q}_u[n]\| \le S_{\max}, n = 1, \ldots, N. \tag{2}$$

Here, we define $S_{\max} \triangleq V_{\max}\phi_t$ as the maximum flight distance that the UAV can reach in one slot.

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B. Transmission and Resource Allocation Model

We make the assumption that the air-to-ground channels among the UAV and the ground users primarily consist of LoS links [38]. According to the free-space path loss model, we describe the channel power gain between the UAV and the kth user during slot n as

$$g_k[n] = h_0 d_k^{-\alpha_L}[n] = \frac{h_0}{\left(\|\mathbf{r}_k - \mathbf{q}_u[n]\|^2 + H^2\right)^{\alpha_L/2}} \tag{3}$$

where h_0 denotes the reference channel power gain, α_L expresses the path loss exponent, and $d_k[n] \triangleq \sqrt{\|\mathbf{q}_u[n] - \mathbf{r}_k\|^2 + H^2}$ is the distance from the UAV to user k during slot n.

Assuming a total system bandwidth of B Hertz (Hz), we define $\zeta_k[n]$ as the fraction of the total bandwidth allocated to user k during slot n. It should be noted that in practical OFDMA systems, $\zeta_k[n]$ is generally a discrete value between 0 and 1, representing the ratio of the number of subcarriers allocated to user k in time slot n. However, when the number of subcarriers is sufficiently large, $\zeta_k[n]$ can be approximated to be a continuous value between 0 and 1 [45]. Therefore, we can express the constraints on bandwidth assignment as

$$0 \le \zeta_k[n] \le 1 \quad \forall k, n \tag{4}$$

$$\sum_{k=1}^{K} \zeta_k[n] \le 1 \quad \forall n. \tag{5}$$

We use $p_k[n]$ to represent the transmission power assigned to the user k by at the slot n, where $p_k[n] \ge 0$ and there is a maximum limit, $P_c^{\max}$, to the total transmission power of the UAV. Thus, the constraints on the power allocation are represented as

$$\sum_{k=1}^{K} p_k[n] \le P_c^{\max} \quad \forall n. \tag{6}$$

The instantaneous transmission rate of user k during slot n is given in bits/second as

$$\begin{aligned} R_k[n] &= \zeta_k[n] B \log_2\left(1 + \frac{p_k[n] g_k[n]}{\zeta_k[n] B N_0}\right) \\ &= \zeta_k[n] B \log_2\left(1 + \frac{p_k[n]\beta_0}{\zeta_k[n]\left(\|\mathbf{q}_u[n] - \mathbf{r}_k\|^2 + H^2\right)^{\alpha_L/2}}\right) \end{aligned} \tag{7}$$

where $\beta_0 \triangleq (h_0/BN_0)$ with N_0 denoting the noise power spectral density. Consequently, the overall transmission rate of kth user over N slots can be expressed as

$$R_k = \sum_{n=1}^{N} R_k[n]. \tag{8}$$

C. UAV Power Consumption Model

The overall power consumption of UAV is mainly composed of communication power consumption and propulsion power consumption [21]. First, the total communication power consumption can be denoted as $P_c[n]$ which is obtained by summing the transmit powers of all users, i.e., $\sum_{k=1}^{K} p_k[n]$, as discussed in Section II-B. Then, the propulsion power consumption is denoted as [46]

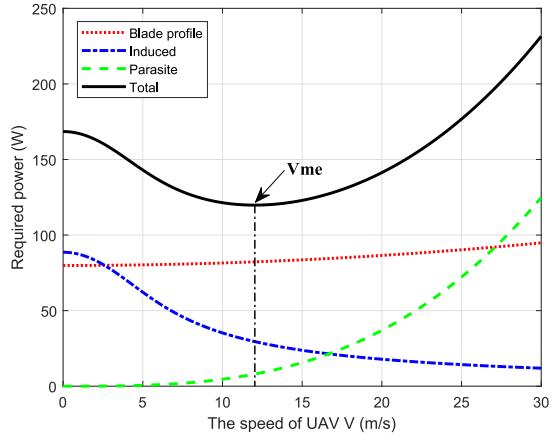

Fig. 2. Propulsion power of the UAV as a function of its speed.

$$P_f[n] = P_{\text{prof}}[n] + P_{\text{indu}}[n] + P_{\text{para}}[n] \tag{9}$$

where

$$P_{\text{prof}}[n] = P_r\left(1 + \frac{3\|\mathbf{v}_u[n]\|^2}{U_t^2}\right) \tag{10}$$

$$P_{\text{indu}}[n] = P_n\left(\sqrt{1 + \frac{\|\mathbf{v}_u[n]\|^4}{4v_z^4}} - \frac{\|\mathbf{v}_u[n]\|^2}{2v_z^2}\right)^{1/2} \tag{11}$$

$$P_{\text{para}}[n] = \frac{1}{2} d_z \rho s A \|\mathbf{v}_u[n]\|^3. \tag{12}$$

The $P_{\text{prof}}[n]$ in (10) denotes the blade profile power where P_r is the constant power value of the UAV in the hovering state and U_t denotes the UAV tip speed about rotor blade. $P_{\text{indu}}[n]$ in (11) expresses the induced power where P_n is the corresponding constant power of the UAV in the hovering state and v_z represents the average induced velocity of the rotor blades during hover. Meanwhile, $P_{\text{para}}[n]$ in (12) is parasite power of UAV where ρ, d_z, A and s represent air density, fuselage drag ratio, rotor disc area and rotor solidity, respectively.

It is worth noting that $\|\mathbf{v}_u[n]\| = \|(\mathbf{q}_u[n-1] - \mathbf{q}_u[n])/\phi_t\|$ represents the UAV speed. We use $\mathbf{v}_u \triangleq \{\mathbf{v}_u[n]\}$ to represent the UAV speed set. The UAV propulsion power, P_f, can be obtained from its speed $V \triangleq \|\mathbf{v}_u[n]\|$, as depicted in Fig. 2. It is worth emphasizing that hovering, i.e., $V = 0$ m/s, is typically not the optimal status in terms of energy-efficient UAV. Instead, in this figure, V_{me} denotes the optimal speed which corresponds to the lowest propulsion power consumption.

Thus, we can denote the overall power consumption of the system across all slots as

$$P_{\text{tot}} = \sum_{n=1}^{N} P_{\text{tot}}[n] = \sum_{n=1}^{N} \left(P_c[n] + P_f[n]\right). \tag{13}$$

D. Problem Formulation

We denote $\mathbf{Q} = \{\mathbf{q}_u[n], n \in \mathcal{N}\}$, $\mathbf{A} = \{\zeta_k[n], k \in \mathcal{K}, n \in \mathcal{N}\}$, $\mathbf{P} = \{p_k[n], k \in \mathcal{K}, n \in \mathcal{N}\}$. In order to prevent a

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winners-take-all scenario and promote fairness, we take the minimum sum rate across all the N time slots of users as the system transmission rate [40]. Hence, the achievable sum rate can be expressed as $\min_{k\in\mathcal{K}} \sum_{n=1}^{N} R_k[n]$, and the system's EE is defined as $EE \triangleq \min_{k\in\mathcal{K}} \sum_{n=1}^{N} R_k[n] / \sum_{n=1}^{N} P_{\text{tot}}[n]$. To maximize the EE of the UAV-assisted downlink communication system while ensuring fairness among users via jointly designing the trajectory $\mathbf{Q}$, the bandwidth allocation $\mathbf{A}$ and the power allocation $\mathbf{P}$, the optimization problem is formulated as

$$\max_{\mathbf{P},\mathbf{A},\mathbf{Q}} \min_{k\in\mathcal{K}} \frac{\sum_{n=1}^{N} R_k[n]}{\sum_{n=1}^{N} P_{\text{tot}}[n]} \tag{14a}$$

$$\text{s.t.} \quad 0 \le \zeta_k[n] \le 1 \quad \forall k, n \tag{14b}$$

$$\sum_{k=1}^{K} \zeta_k[n] \le 1 \quad \forall n \tag{14c}$$

$$p_k[n] \ge 0 \quad \forall k, n \tag{14d}$$

$$\sum_{k=1}^{K} p_k[n] \le P_c^{\max} \quad \forall n \tag{14e}$$

$$\|\mathbf{q}_u[n-1] - \mathbf{q}_u[n]\| \le S_{\max} \quad \forall n \tag{14f}$$

$$q[0] = q[N] \tag{14g}$$

$$\mathbf{v}_u[n] = (\mathbf{q}_u[n] - \mathbf{q}_u[n-1])/\phi_t. \tag{14h}$$

Constraints (14b) and (14c) denote the limitations on bandwidth allocation, (14d) and (14e) denote the constraints on power allocation, and (14f)–(14h) give the limitations to the UAV flight trajectory.

Due to the nonsmooth nature of $\min_{k\in\mathcal{K}} \sum_{n=1}^{N} R_k[n]$ in (14a), we introduce $\lambda \triangleq \min_{k\in\mathcal{K}} \sum_{n=1}^{N} R_k[n]$. This transformation allows us to convert the original problem (14) into a smooth problem

$$\max_{\lambda,\mathbf{P},\mathbf{A},\mathbf{Q}} \frac{\lambda}{\sum_{n=1}^{N} P_{\text{tot}}[n]} \tag{15a}$$

$$\text{s.t.} \quad \sum_{n=1}^{N} R_k[n] \ge \lambda \quad \forall k$$
$$(14b)-(14h). \tag{15b}$$

It is evident that the objective (15a) can be observed as a fractional equation composed of a linear numerator and a nonconvex denominator, while (15b) is nonconvex. Consequently, (15) is a challenging nonconvex fractional program problem that is difficult to solve directly.

III. Proposed Algorithm

In this section, we propose an efficient iterative algorithm to solve the formulated problem (15) by adopting the BCD framework [47]. The proposed algorithm involves first optimizing the power and bandwidth assignment $\{\mathbf{P}, \mathbf{A}\}$ using Dinkelbach's algorithm and Lagrange dual technique, given a trajectory of UAV $\mathbf{Q}$. Similarly, we then optimize the trajectory design of UAV $\mathbf{Q}$ via the SCA method and Dinkelbach's algorithm, given a set of power and bandwidth assignment $\{\mathbf{P}, \mathbf{A}\}$. The two subproblems are optimized and updated in an alternating manner till convergence. Then, we outline the comprehensive algorithm and the convergence and complexity analysis of the algorithm are given. Finally, we introduce a trajectory initialization scheme of UAV with low-complexity.

A. Power and Bandwidth Allocation of UAV

To begin, we focus on the joint optimization of $\{\mathbf{P}, \mathbf{A}\}$ while assuming that the trajectory of UAV $\mathbf{Q}$ is given. By doing so, we transform the optimization problem (15) into

$$\max_{\lambda,\mathbf{P},\mathbf{A}} \frac{\lambda}{\sum_{n=1}^{N}\left(\sum_{k=1}^{K} p_k[n] + P_f[n]\right)} \tag{16a}$$

$$\text{s.t.} \quad 0 \le \zeta_k[n] \le 1 \quad \forall k, n \tag{16b}$$

$$\sum_{k=1}^{K} \zeta_k[n] \le 1 \quad \forall n \tag{16c}$$

$$p_k[n] \ge 0 \quad \forall k, n \tag{16d}$$

$$\sum_{k=1}^{K} p_k[n] \le P_c^{\max} \quad \forall n \tag{16e}$$

$$\sum_{n=1}^{N} \zeta_k[n] B \log_2\left(1 + \frac{p_k[n] H_k[n]}{\zeta_k[n]}\right) \ge \lambda \quad \forall k \tag{16f}$$

where $H_k[n] \triangleq (g_k[n]/BN_0)$. When $\zeta_k[n]$ equals 0, we define $\zeta_k[n] B \log_2(1 + [p_k[n] H_k[n]]/[\zeta_k[n]]) \triangleq 0$.

Proposition 1: Problem (16) is a quasi-convex fractional optimization problem.

Proof: See the Appendix. ■

With Proposition 1, the fractional maximization problem (16) can be solved optimally and efficiently via Dinkelbach's method [41]. For this purpose, we transform the fractional objective function in (16) into a tractable subtraction form as

$$f(\tau^m) \triangleq \max_{\lambda,\mathbf{P},\mathbf{A}} \lambda - \tau^{(m)} s(\mathbf{P}, \mathbf{A}) \tag{17}$$

where $s(\mathbf{P}, \mathbf{A}) \triangleq (\sum_{n=1}^{N}(\sum_{k=1}^{K} p_k[n] + P_f[n]))$ denotes the total energy consumption function with respect to $\{\mathbf{P}, \mathbf{A}\}$ and $\tau^{(m)}$ denotes the optimal system EE in the mth iteration acquired via solving the following convex problem:

$$\max_{\lambda,\mathbf{P},\mathbf{A}} \lambda - \tau^{(m)} s(\mathbf{P}, \mathbf{A})$$
$$\text{s.t.} \quad (16b)-(16f). \tag{18a}$$

In order to get the optimal solution of (16), we need to iteratively optimize (18). Thus, we propose an EE maximization iterative algorithm by applying Dinkelbach's method. The details of the proposed algorithm are shown in Algorithm 1.

It is worth noting that the optimization problem (18) needed to be solved in step 3 of Algorithm 1 is convex. It can be optimally solved using the convex optimization solver, CVX that exploits the interior point method [48]. In order to look at the internal structure of the optimal solution for power and bandwidth allocation, we apply the Lagrange duality method [45]. This is because that the fulfillment of Slater's constraint has been established [49], thereby ensuring the

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Algorithm 1 Proposed Optimization Algorithm Based on Dinkelbach's Method for Solving Problem (16)

Initialize m=0 and $\tau^{(m)}$, the error tolerance Δ_1, the maximum iteration number M.
while $m < M$ **do**
  Solve the optimization problem (18) with given $\tau^{(m)}$ and obtain resource allocation policy $\{\lambda^{m+1}, \mathbf{P}^{m+1}, \mathbf{A}^{m+1}\}$.
  if $\lambda^{m+1} - \tau^{(m)} s(\mathbf{P}, \mathbf{A}) < \Delta_1$ **then**
    Return $\{\lambda^*, \mathbf{P}^*, \mathbf{A}^*\} = \{\lambda^{m+1}, \mathbf{P}^{m+1}, \mathbf{A}^{m+1}\}$ and $\tau^{(*)} = \lambda^*/s(\mathbf{P}^*, \mathbf{A}^*)$.
  else
    $\tau^{(m+1)} = \lambda^{m+1}/s(\mathbf{P}^{m+1}, \mathbf{A}^{m+1})$.
    Update $m = m + 1$.
  end if
end while
Obtain the optimal power and bandwidth allocation police $\{\mathbf{P}^*, \mathbf{A}^*\}$.

validity of strong duality. Expressing (18) in terms of the Lagrange function is as follows:

$$\begin{aligned}\mathcal{L}(\lambda, \mathbf{P}, \mathbf{A}, \Omega, \Theta, \Psi) &= \lambda - \tau^{(m)}\left(\sum_{n=1}^{N}\left(\sum_{k=1}^{K} p_k[n] + P_f[n]\right)\right) \\ &+ \sum_{n=1}^{N}\psi_n\left(P_{\max} - \sum_{k=1}^{K} p_k[n]\right) + \sum_{n=1}^{N}\theta_n\left(1 - \sum_{k=1}^{K}\zeta_k[n]\right) \\ &+ \sum_{k=1}^{K}\omega_k\left(\sum_{n=1}^{N}\zeta_k[n]B\log_2\left(1 + \frac{p_k[n]H_k[n]}{\zeta_k[n]}\right) - \lambda\right) \\ &= \left(1 - \sum_{k=1}^{K}\omega_k\right)\lambda - \sum_{n=1}^{N}\sum_{k=1}^{K}\left(\tau^{(m)} + \psi_n\right)p_k[n] \\ &+ \sum_{k=1}^{K}\sum_{n=1}^{N}\omega_k\zeta_k[n]B\log_2\left(1 + \frac{p_k[n]H_k[n]}{\zeta_k[n]}\right) + \sum_{n=1}^{N}\theta_n \\ &- \sum_{k=1}^{K}\sum_{n=1}^{N}\theta_n\zeta_k[n] - \tau^{(m)}\sum_{n=1}^{N}P_f[n] + \sum_{n=1}^{N}\psi_n P_{\max} \end{aligned} \tag{19}$$

where $\Omega = \{\omega_k \ \forall k\}$, $\Theta = \{\theta_n \ \forall n\}$, $\Psi = \{\psi_n \ \forall n\}$ denote those nonnegative vectors of Lagrange multipliers. Consequently, the dual function can be formulated as

$$\begin{aligned} f(\Omega, \Theta, \Psi) = \max_{\lambda, \mathbf{P}, \mathbf{A}} \ & \mathcal{L}(\lambda, \mathbf{P}, \mathbf{A}, \Omega, \Theta, \Psi) \\ \text{s.t.} \ & (16b), \ (16d). \end{aligned} \tag{20a}$$

Based on that, we introduce the following lemma.

Lemma 1: In order to ensure the boundedness of $f(\Omega, \Theta, \Psi)$ in (20), it is necessary for the condition $1 - \sum_{k=1}^{K}\omega_k = 0$ to be satisfied.

Proof: The proof can be established through a contradiction. If $1 - \sum_{k=1}^{K}\omega_k < 0$ or $1 - \sum_{k=1}^{K}\omega_k > 0$, it would result in $f(\Omega, \Theta, \Psi) \to +\infty$ when $\lambda \to -\infty$ or $\lambda \to +\infty$. Hence, neither of the aforementioned inequalities is true, thus confirming the validity of the lemma. ■

Thus, we formulate the dual problem as

$$\min_{\Omega, \Theta, \Psi} \max_{\lambda, \mathbf{P}, \mathbf{A}} \quad \mathcal{L}(\lambda, \mathbf{P}, \mathbf{A}, \Omega, \Theta, \Psi) \tag{21a}$$

$$\text{s.t.} \ \ 1 - \sum_{k=1}^{K}\omega_k = 0 \tag{21b}$$

$$\Omega \succeq 0, \ \Theta \succeq 0, \ \Psi \succeq 0. \tag{21c}$$

The dual problem provided can be solved through iteratively address two nested loops, i.e., the inner loop which maximizes $\{\lambda, \mathbf{P}, \mathbf{A}\}$ with given $\{\Omega, \Theta, \Psi\}$, and the outer loop which serves as the master dual problem minimizing $\{\Omega, \Theta, \Psi\}$. In the following, we give the algorithm details.

1) Inner Loop: Given the dual variables Ω, Θ, Ψ, (20) can be divided into $KN+1$ subproblems that can be independently and concurrently addressed. These subproblems are as follows:

$$\max_{\lambda} \quad \left(1 - \sum_{k=1}^{K}\omega_k\right)\lambda \tag{22}$$

$$\begin{aligned} \max_{p_k[n], \zeta_k[n]} \ & \omega_k\zeta_k[n]B\log_2\left(1 + \frac{p_k[n]H_k[n]}{\zeta_k[n]}\right) - \theta_n\zeta_k[n] \\ & - \left(\tau^{(m)} + \psi_n\right)p_k[n] \end{aligned} \tag{23a}$$

$$\text{s.t.} \ \ 0 \le \zeta_k[n] \le 1 \quad \forall k, n \tag{23b}$$

$$p_k[n] \ge 0 \quad \forall k, n \tag{23c}$$

where (23) is one of the KN subproblems for user k and time slot n. Based on Lemma 1, the optimal function value is 0 in (22), no matter what value λ takes. For simplicity, we set the optimal solution $\lambda^\star = 0$.

Then, because the objective in (23) is jointly concave for $p_k[n]$ and $\zeta_k[n]$, we apply the Karush–Kuhn–Tucker (KKT) conditions to solve it. By differentiating (23a) over $p_k[n]$, we obtain the corresponding KKT condition as

$$\omega_k B\frac{H_k[n]\zeta_k[n]}{(\zeta_k[n] + p_k[n]H_k[n])\log(2)} - \psi_n - \tau^{(m)} = 0. \tag{24}$$

Thus, we can express the optimal solution $p_k^\star[n]$ as

$$p_k^\star[n] = \zeta_k[n]\left[\frac{\omega_k B}{\left(\psi_n + \tau^{(m)}\right)\log(2)} - \frac{1}{H_k[n]}\right]^+ \tag{25}$$

where $[y]^+$ means taking the maximum between y and 0. It is important to note that the strategy in (25) adopts a multilevel water-filling framework. We substitute it into (23) and have

$$\max_{\zeta_k[n]} \ \zeta_k[n]f\left(\hat{p}_k[n]\right) \tag{26a}$$

$$\text{s.t.} \ \ 0 \le \zeta_k[n] \le 1 \quad \forall k, n \tag{26b}$$

where $\hat{p}_k[n] \triangleq [(p_k^\star[n])/(\zeta_k[n])] = [\omega_k B/[(\psi_n + \tau^{(m)})\log(2)] - (1/H_k[n])]^+$ and $f(\hat{p}_k[n]) \triangleq \omega_k B\log_2(1 + \hat{p}_k[n]H_k[n]) - (\tau^{(m)} + \psi_n)\hat{p}_k[n] - \theta_n$. We find (26) is a linear program (LP). Through the analysis, we can get the optimal bandwidth allocation $\zeta_k^\star[n]$, which is a piece-wise function as

$$\zeta_k^\star[n] = \begin{cases} 1, \ \text{for} \ f\left(\hat{p}_k[n]\right) > 0 \\ c, \ \text{for} \ f\left(\hat{p}_k[n]\right) = 0 \\ 0, \ \text{for} \ f\left(\hat{p}_k[n]\right) < 0 \end{cases} \tag{27}$$

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where c can take arbitrary value between 0 and 1 because $f(\hat{p}_k[n]) = 0$ will make the objective function value equal to 0 no matter what value c takes. As the same as λ, the c is simply set to zero.

2) Outer Loop: Based on the obtained solutions $\lambda^\star$, $\mathbf{A}^\star$, $\mathbf{P}^\star$ to problems (22) and (23), we obtain the dual function $f(\Omega, \Theta, \Psi)$. Then, (21) needs to be solved to get the optimal dual variables. The common subgradient based method can be applied to optimize this problem. We use the subgradients of (21a) and (21b) and (21c) of (21) to optimize the dual variables Ω, Θ, Ψ. First, the subgradient of (21a) can be expressed as $e_0 = [\Delta\Omega^T, \Delta\Theta^T, \Delta\Psi^T]^T$, where $\Delta\Omega$, $\Delta\Theta$, and $\Delta\Psi$ are vectors stacked by the following variables, respectively:

$$\Delta\omega_k = \sum_{n=1}^{N} \zeta_k[n] B \log_2\left(1 + \frac{p_k[n]H_k[n]}{\zeta_k[n]}\right) \quad \forall k \tag{28}$$

$$\Delta\theta_n = 1 - \sum_{k=1}^{K} \zeta_k[n] \quad \forall n \tag{29}$$

$$\Delta\psi_n = P_{\max} - \sum_{k=1}^{K} p_k[n] \quad \forall n. \tag{30}$$

Then, for (21b), we use two inequality constraints to equivalently transform it. They are $1 - \sum_{k=1}^{K} \omega_k \leq 0$ and $1 - \sum_{k=1}^{K} \omega_k \geq 0$, respectively. The subgradient for the former is $e_1 = [\Delta\Omega^T, \Delta\Theta^T, \Delta\Psi^T]^T$, where $\Delta\omega_k = -1$, $\Delta\theta_n = 0$, and $\Delta\psi_n = 0$. While the subgradient for the latter is $e_2 = -e_1$. Based on these obtained subgradients, the ellipsoid method with constraints can be used to update the dual variables Ω, Θ, Ψ to obtain the optimal solution [45]. Until here, the outer loop ends.

It is important to note that for the convex optimization problem, one solution that maximizes the Lagrangian function corresponds to the optimal primal solution only if this solution is both unique and primal feasible [45]. But we can find the optimal solution $\lambda^\star$ and $\mathbf{A}^\star$ for maximizing $\mathcal{L}(\lambda, \mathbf{P}, \mathbf{A}, \Omega, \Theta, \Psi)$ are not unique when $1 - \sum_{k=1}^{K} \omega_k = 0$ and $f(\hat{p}_k[n]) = 0$ as mentioned above. Thus, we need additional operations to obtain the optimal solution. Specially, it can be found that based on the above optimal dual variables Ω^*, Θ^*, Ψ^*, the $\hat{p}_k[n]$ which can be obtained from (25) is unique. So we substitute $\hat{p}_k[n]$ into (18) to obtain

$$\max_{\lambda, \mathbf{A}} \ \lambda - \tau^{(m)}\left(\sum_{n=1}^{N}\left(\sum_{k=1}^{K} \alpha_k[n]\hat{p}_k[n] + P_f[n]\right)\right) \tag{31a}$$

$$\text{s.t.} \ \ 0 \leq \zeta_k[n] \leq 1 \quad \forall k, n \tag{31b}$$

$$\sum_{k=1}^{K} \zeta_k[n] \leq 1 \quad \forall n \tag{31c}$$

$$\zeta_k[n]\hat{p}_k[n] \geq 0 \quad \forall k, n \tag{31d}$$

$$\sum_{k=1}^{K} \zeta_k[n]\hat{p}_k[n] \leq P_c^{\max} \quad \forall n \tag{31e}$$

$$\sum_{n=1}^{N} \zeta_k[n] B \log_2\big(1 + \hat{p}_k[n]H_k[n]\big) \geq \lambda \quad \forall k. \tag{31f}$$

Algorithm 2 Optimal Resource Allocation Algorithm for Solving Problem (18)

1: Initial the dual variables Ω, Θ, Ψ, the ellipsoid, the error tolerance Δ_2.
2: **repeat**
3: Solving problems (22) and (23) to obtain $\lambda^\star$, $\mathbf{A}^\star$, $\mathbf{P}^\star$.
4: Get the subgradients of (21a) and the constraints (21b) and (21c) of problem (21).
5: Optimize Ω, Θ, Ψ via applying the ellipsoid method with constrained.
6: **until** Ω, Θ, Ψ convergence within Δ_2.
7: Get the optimal dual variables $\{\Omega^*, \Theta^*, \Psi^*\}$.
8: Obtain the optimal resource allocation scheme $\{\lambda^*, \mathbf{A}^*, \mathbf{P}^*\}$ through solving problem (31).

Now, the optimization problem (31) can be categorized as an LP that can be addressed using standard convex optimization tool like the *linprog* function in MATLAB [50]. This function incorporates various algorithms like the interior point method and the simplex algorithm. After obtaining the optimal solution about bandwidth allocation, $\mathbf{A}^*$, the optimal solution of power allocation $\mathbf{P}^*$ will be determined where each element is denoted as $p_k^*[n] = \zeta_k^*[n]\hat{p}_k^*[n]$. Therefore, the specific steps involved in obtaining the optimal resource allocation of (18) outlined in step 3 of Algorithm 1 are summarized in Algorithm 2.

B. Trajectory Design of UAV

Given the feasible resource allocation scheme $\{\mathbf{P}, \mathbf{A}\}$, the optimal problem (15) will be formulated to optimize the UAV trajectory $\mathbf{Q}$ as

$$\max_{\lambda, \mathbf{Q}} \ \frac{\lambda}{\sum_{n=1}^{N} P_{\text{tot}}[n]} \tag{32a}$$

$$\text{s.t.} \ \ \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|^2 \leq S_{\max}^2 \quad \forall n \tag{32b}$$

$$q[0] = q[N] \tag{32c}$$

$$\mathbf{v}_u[n] = (\mathbf{q}_u[n] - \mathbf{q}_u[n-1])/\delta_t \tag{32d}$$

$$\sum_{n=1}^{N} \zeta_k[n] B \log_2\left(1 + \frac{A_k[n]}{\left(\|\mathbf{q}_u[n] - \mathbf{r}_k\|^2 + H^2\right)^{\alpha_L/2}}\right) \geq \lambda \quad \forall k \tag{32e}$$

where $A_k[n] \triangleq (p_k[n]\beta_0/\alpha_k[n])$. Problem (32) is nonconvex because of the nonconvex nature of (32a) and (32e). To solve it, we introduce a new variable set $\mathbf{U} = \{U_k[n] \ \ \forall k \in \mathcal{K}, n \in \mathcal{N}\}$, where

$$\frac{1}{\left(\|\mathbf{q}_u[n] - \mathbf{r}_k\|^2 + H^2\right)^{\alpha_L/2}} \geq U_k[n] \quad \forall k, n. \tag{33}$$

Next, we introduce a new variable set $\mathbf{X} = \{x[n] | x[n] \geq 0 \ \ \forall n \in \mathcal{N}\}$, in which

$$x[n] \geq \left(\sqrt{1 + \frac{\|\mathbf{v}_u[n]\|^4}{4v_z^4}} - \frac{\|\mathbf{v}_u[n]\|^2}{2v_z^2}\right)^{1/2} \tag{34}$$

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that can be equivalently expressed as

$$\frac{1}{x^2[n]} \leq x^2[n] + \frac{\|\mathbf{v}_u[n]\|^2}{v_z^2}. \tag{35}$$

Then, we transform (32) into

$$\max_{\lambda,\mathbf{Q},\mathbf{U},\mathbf{X}} \quad \frac{\lambda}{\sum_{n=1}^{N}\left(\tilde{P}_f[n] + \sum_{k=1}^{K} p_k[n]\right)} \tag{36a}$$

$$\text{s.t.} \quad \sum_{n=1}^{N} \zeta_k[n] B \log_2(1 + A_k[n] U_k[n]) \geq \lambda \tag{36b}$$

$$\frac{1}{\left(\|\mathbf{q}_u[n] - \mathbf{r}_k\|^2 + H^2\right)^{\alpha_L/2}} \geq U_k[n] \quad \forall k, n \tag{36c}$$

$$x[n] \geq 0 \quad \forall n \tag{36d}$$

$$\frac{1}{x^2[n]} \leq x^2[n] + \frac{\|\mathbf{v}_u[n]\|^2}{v_z^2} \quad \forall n$$

$$(32b)-(32d) \tag{36e}$$

where

$$\tilde{P}_f[n] \triangleq P_r\left(1 + \frac{3\|\mathbf{v}_u[n]\|^2}{U_t^2}\right) + P_n x[n] + \frac{1}{2} d_z \rho s A \|\mathbf{v}_u[n]\|^3. \tag{37}$$

Now, (36) remains nonconvex due to the presence of (36c) and (36e). To address that, we adopt the SCA, which allows us to iteratively get a suboptimal solution [51]. Specifically, we find that the left side of (36c) keeps convex in terms of $(\|\mathbf{q}_u[n] - \mathbf{r}_k\|^2 + H_0^2)$ with $\alpha_L \geq 2$. Thus, we utilize $\mathbf{Q}^l = \{\mathbf{q}_u^l[n]\}$ to represent the UAV trajectory got at the lth iteration, then we acquire the following lower bound as:

$$\begin{aligned}&\frac{1}{\left(\|\mathbf{r}_k - \mathbf{q}_u[n]\|^2 + H^2\right)^{\alpha_L/2}}\\ &\geq B_k^l[n] - C_k^l[n]\left(\|\mathbf{r}_k - \mathbf{q}_u[n]\|^2 - \|\mathbf{r}_k - \mathbf{q}_u^l[n]\|^2\right)\\ &\triangleq D_k^{lb,r}[n]\end{aligned} \tag{38}$$

where

$$B_k^l[n] = \frac{1}{\left(\|\mathbf{r}_k - \mathbf{q}_u^l[n]\|^2 + H^2\right)^{\alpha_L/2}} \tag{39}$$

and

$$C_k^l[n] = \frac{\alpha_L/2}{\left(\|\mathbf{r}_k - \mathbf{q}_u^l[n]\|^2 + H^2\right)^{(\alpha_L/2+1)}}. \tag{40}$$

Then, for (36e), it is observed that $x^2[n] + \|\mathbf{v}_u[n]\|^2/v_z^2$ is jointly convex with respect to $\mathbf{v}_u[n]$ and $x[n]$. By utilizing the given localized point at the lth iteration $\mathbf{v}_u^l[n]$ and $x^l[n]$, its lower bound can be derived utilizing SCA, which is

$$\begin{aligned}&\frac{\|\mathbf{v}_u[n]\|^2}{v_z^2} + x^2[n] \geq \left(x^l[n]\right)^2 + 2x^l[n]\left(x[n] - x^l[n]\right)\\ &\quad + \frac{\|\mathbf{v}_u^l[n]\|^2}{v_z^2} + \frac{2\left(\mathbf{v}_u^l[n]\right)^T}{v_z^2}\left(\mathbf{v}_u[n] - \mathbf{v}_u^l[n]\right) \triangleq E^{lb,l}[n].\end{aligned} \tag{41}$$

Hence, utilizing the localized point at the lth iteration of the auxiliary variable and trajectory variable $\{\mathbf{X}^l, \mathbf{Q}^l\}$, along with the lower bounds presented at (38) and (41), (36) can be formulated as

$$\max_{\lambda,\mathbf{Q},\mathbf{U},\mathbf{X}} \quad \frac{\lambda}{\sum_{n=1}^{N}\left(\tilde{P}_f[n] + \sum_{k=1}^{K} p_k[n]\right)} \tag{42a}$$

$$\text{s.t.} \quad U_k[n] \leq D_k^{lb,r}[n] \quad \forall k, n \tag{42b}$$

$$\frac{1}{x^2[n]} \leq F^{lb,r}[n] \quad \forall n$$

$$(32b)-(32d),\ (36b),\ (36d). \tag{42c}$$

Now, (42) becomes a convex fractional optimization problem, that can be optimally solved using Dinkelbach's method in conjunction with CVX solvers [48], [52]. The algorithmic procedure is similar with that of Algorithm 1, which is therefore omitted here. It is worth noting that the application of lower bounds in (38) as well as (41) indicates that the feasible solution space for (42) is encompassed within that of (32), i.e., the optimal objective function value of (42) is a lower bound of that of (32).

Algorithm 3 BCD Iterative Algorithm for Solving Problem (15)

1: Set the iterative index $l = 0$, the starting trajectory of UAV $\mathbf{Q}^0$, the maximum tolerance Δ_3.
2: **repeat**
3: Optimize the quasi-convex subproblem (16) with $\{\mathbf{Q}^l\}$ by using **Algorithm 1** and **Algorithm 2** and get the optimal resource allocation solutions $\{\mathbf{P}^{l+1}, \mathbf{A}^{l+1}\}$.
4: Optimize the subproblem (42) with $\{\mathbf{Q}^l, \mathbf{P}^{l+1}, \mathbf{A}^{l+1}\}$, update the trajectory solution of UAV as $\{\mathbf{Q}^{l+1}\}$.
5: Set $l = l + 1$.
6: **until** The difference of the two consecutive objective values is smaller than the maximum tolerance Δ_3.
7: Output optimal solution of power allocation, bandwidth allocation and trajectory optimization as $\{\mathbf{P}^*, \mathbf{A}^*, \mathbf{Q}^*\} = \{\mathbf{P}^l, \mathbf{A}^l, \mathbf{Q}^l\}$.

C. Convergence and Complexity of the Overall Algorithm

After solving the subproblems in the above two sections, respectively, we propose an efficient iterative approach, employing the BCD framework, to tackle (15). We iteratively solve out the resource allocation $\{\mathbf{A}, \mathbf{P}\}$ and trajectory optimization $\mathbf{Q}$ to get an efficient suboptimal solution. The specific steps of the overall algorithm is summarized in Algorithm 3.

1) Convergence of the BCD Method: The convergence property of the proposed algorithm is analyzed. To begin, let us denote $\{\mathbf{P}^l, \mathbf{A}^l\}$ and $\mathbf{Q}^l$ as the results obtained via solving the power and bandwidth allocation subproblem and trajectory optimization subproblem in iteration l, subsequently, we can derive the following expression in the resource allocation block as:

$$\begin{aligned}&\gamma\left(\mathbf{P}^{l-1}, \mathbf{A}^{l-1}, \mathbf{Q}^{l-1}\right) = \gamma_{PA}\left(\mathbf{P}^{l-1}, \mathbf{A}^{l-1}, \mathbf{Q}^{l-1}\right)\\ &\leq \gamma_{PA}\left(\mathbf{P}^{l}, \mathbf{A}^{l}, \mathbf{Q}^{l-1}\right) = \gamma\left(\mathbf{P}^{l}, \mathbf{A}^{l}, \mathbf{Q}^{l-1}\right)\end{aligned} \tag{43}$$

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where $\gamma(\cdot)$ and $\gamma_{PA}(\cdot)$ represent the system EE obtained from (15a) and (16a), respectively. The inequality holds because the optimal solution is found for the resource allocation subproblem.

While optimizing the UAV trajectory, we utilize $\gamma_Q^{lb,r}(\cdot)$ to denote the system EE obtained from (42a). Similar to the analysis in [21], we get

$$\gamma\left(\mathbf{P}^l, \mathbf{A}^l, \mathbf{Q}^{l-1}\right) = \gamma_Q^{lb,l}\left(\mathbf{P}^l, \mathbf{A}^l, \mathbf{Q}^{l-1}\right) \leq \gamma_Q^{lb,r}\left(\mathbf{P}^l, \mathbf{A}^l, \mathbf{Q}^l\right) \leq \gamma\left(\mathbf{P}^l, \mathbf{A}^l, \mathbf{Q}^l\right). \tag{44}$$

Combining (43) and (44), we have

$$\gamma\left(\mathbf{P}^{l-1}, \mathbf{A}^{l-1}, \mathbf{Q}^{l-1}\right) \leq \gamma\left(\mathbf{P}^l, \mathbf{A}^l, \mathbf{Q}^l\right) \tag{45}$$

which showing that the optimized objective value got in (15a) of (15) does not decrease along with each iteration. Meanwhile, the value is bounded by a positive finite number. Thus, we concluded that Algorithm 3 is convergent.

2) Complexity of the BCD Method: In this section, we analyze the computational complexity of the designed overall algorithm, i.e., Algorithm 3. Its complexity consists mainly of solving subproblem (16) using Algorithms 1 and 2 in step 3 and solving subproblem (42) in step 4. Since Algorithm 2 is included in step 3 of Algorithm 1, we first analyze the complexity of Algorithm 2. The complexity of Algorithm 2 is composed of three parts. The first part is solving (22) and (23) in step 3, whose complexities are $\mathcal{O}(1)$ and $\mathcal{O}(KN)$, respectively. The second part is from that optimizing the dual variables through the ellipsoid method in step 5 whose complexity is $\mathcal{O}(N^2)$. As this method requires $\mathcal{O}(N^2)$ iterations to converge, steps 2 to 6 has the overall complexity $\mathcal{O}(N^4)$ [45]. In addition, the third part is from optimizing (31), which has polynomial-level complexity due to the use of the *linprog* function. Therefore, the total complexity of Algorithm 2 is denoted as $\mathcal{O}(N^4)$. Consequently, the complexity of Algorithm 1 for addressing (16), which consists Algorithm 2 in step 3, can be expressed as $\mathcal{O}(I_1 N^4)$. I_1 represents the number of iterations in Dinkelbach's method. Furthermore, the complexity for solving (42) using the CVX tool and Dinkelbach's algorithm is $\mathcal{O}(I_2 K^{3.5} N^{3.5})$, where I_2 is also a number of iteration. In summary, the overall complexity of Algorithm 3 can be expressed as $\mathcal{O}(I'(I_1 N^4 + I_2 K^{3.5} N^{3.5}) \log(1/\Delta_3))$, where I' represents the number of iterations in BCD algorithm.

D. Trajectory Initialization

Note that for iterative algorithms, the initialization is of great importance for the convergence as well as the system performance. Therefore, the initial scheme $\mathbf{Q}^0 = \{\mathbf{q}_u^0[n] \quad \forall n\}$ for trajectory design needs to be designed properly. Given that the UAV needs to return to its starting point at the end of the service period, and in order to sequentially serve all users randomly distributed in this area, it is intuitively reasonable to apply a more easily created circular trajectory as the initialized trajectory [53]. Choosing that, the center $\mathbf{n}_u = [x_u, y_u]$ and radius c_u of this circle are needed to be designed. For convenience, we simply set the center of this circle to be the center of the service area, which is denoted by $\mathbf{n}_u = [0, 0]$.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Radius of circular cell R	1500 m
Maximum speed of UAV $V_{\max}$	50 m/s
Maximum UAV transmit power limit P_{max}	1 W
Flight altitude of UAV H	300 m
Flight period T	200 s
Path loss exponent α_L	2.5
Reference channel power gain h_0	-50 dB
Total system bandwidth B	1 MHz
Noise power spectral density N_0	-169 dBm/Hz

Next, we decide the radius of this circle through two aspects. For the first aspect, with the UAV speed limit $V_{\max}$, the radius for the circle using the furthest flyable distance of the UAV as its circumference can be denoted by

$$c_m = \frac{V_{\max} T}{2\pi}. \tag{46}$$

For the second aspect, considering balancing the distance of the center user and the edge users from the circular trajectory of the UAV, we first determine the radius formed by the user farthest from the center, which is expressed as

$$c_e = \max_{k \in \mathcal{K}} \|\mathbf{r}_k - \mathbf{n}_u\|. \tag{47}$$

Then, we pick half, i.e., $c_e' = (c_e/2)$, as the second radius selection criterion to balance the users inside and outside the circle.

Combining the above two aspects, we can determine the radius of the initial trajectory as $c_u \triangleq \min\{c_m, c_e'\}$. Thus, with the radius c_u, center $\mathbf{n}_u$, and the arc $\sigma_n = 2\pi(n-1/N-1)$ for all n, we can characterize the initial trajectory $\mathbf{Q}^0$ as follows:

$$\mathbf{q}_u^0[n] = (x_u + c_u \cos\sigma_n, y_u + c_u \sin\sigma_n) \quad \forall n \in \mathcal{N}. \tag{48}$$

IV. SIMULATION RESULTS

In this section, we conduct extensive simulations to verify the performance of the proposed algorithms. The parameters related to the propulsion power consumption of the rotary-wing UAV, i.e., P_n, P_r, U_t, v_z, A, s, d_z and ρ, are set to be the same as those in [38]. Within the considered system, six ground users are randomly distributed in the service area and their locations are marked by '*' as shown in Fig. 4. The rest parameters are listed in Table I. To validate the advantages of the proposed scheme, named as *J-APQ-maxEE*, four benchmark schemes are simulated for comparisons, which are described as follows.

1) *J-AP-CIRCLE:* Scheme which only optimizes power and bandwidth allocation. The UAV trajectory is designed based on our proposed initialization scheme.
2) *J-AP-STATIC:* Scheme which only optimizes power and bandwidth allocation. The UAV statically hovers at the starting point over the whole service period.

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TABLE II
COMPARISON OF COMPUTATIONAL COMPLEXITY

Simulated Algorithms	Computational Complexity
J-APQ-maxEE	$\mathcal{O}(I'(I_1N^4 + I_2K^{3.5}N^{3.5})\log(1/\Delta_3))$
J-AP-CIRCLE	$\mathcal{O}(I_1^cN^4)$
J-AP-STATIC	$\mathcal{O}(I_1^sN^4)$
STATIC	$\mathcal{O}(1)$
J-APQ-maxR	$\mathcal{O}(I_r'(N^4 + K^{3.5}N^{3.5})\log(1/\Delta_3))$
NO-FAIRNESS	$\mathcal{O}(I_1^n(I_n'(N^4 + K^{3.5}N^{3.5})\log(1/\Delta_3)))$

3) *STATIC:* Scheme whose UAV is static, and the power and bandwidth are equally allocated.
4) *J-APQ-maxR [45]:* Scheme which jointly optimizes power allocation, bandwidth allocation and trajectory design, but the propulsion power consumption of the UAV is not taken into account.
5) *NO-FAIRNESS [37]:* Scheme which maximizes the EE regardless of user fairness. Each user has a minimum Quality of Service (QoS) threshold with respect to the achievable rate.

We first show the complexity of all the schemes in Table II. The overall complexity of the proposed J-APQ-maxEE comes from the previous analysis in Section III-C2. The complexity of J-AP-CIRCLE and J-AP-STATIC only comes from optimizing the joint power and bandwidth allocation subproblem, with I_1^c and I_1^s representing the number of iterations in Dinkelbach's method. Because the UAV trajectory, power and bandwidth allocation are all known in advance in STATIC, so its complexity is denoted as $\mathcal{O}(1)$. The computational complexity of J-APQ-maxR comes mainly from applying BCD algorithm to iteratively solve power and bandwidth allocation subproblem and trajectory design subproblem. In which, I_r' represents the iteration number of the BCD algorithm. The complexity of NO-FAIRNESS comes mainly from using Dinkelbach's algorithm to process the fractional objective and using BCD to iteratively solve the power and bandwidth allocation subproblem and the trajectory design subproblem, where I_1^n and I_n' represent the number of iterations in Dinkelbach's algorithm and BCD algorithm, respectively.

To characterize the user fairness, we adopt Jain's fairness index (JFI) [54] to measure the fairness of the user achievable rate, which is denoted as

$$\text{JFI} = \frac{\left(\sum_{k=1}^{K}\left(\sum_{n=1}^{N} R_k[n]\right)\right)^2}{K\left(\sum_{k=1}^{K}\left(\sum_{n=1}^{N} R_k[n]\right)^2\right)} \tag{49}$$

where JFI $\in [0, 1]$. A smaller JFI indicates a larger difference between users' achievable rates ($\sum_{n=1}^{N} R_k[n] \quad \forall k$). The larger the JFI is, the higher the user fairness will be. When all users have equal rates, JFI $= 1$.

In addition to JFI, we have introduced another calculation method of fairness, named as root squared deviation (RSD) [55], which can be expressed as

$$\text{RSD} = \sqrt{\sum_{k=1}^{K}\left(\sum_{n=1}^{N} R_k[n] - \overline{R_k}\right)^2}. \tag{50}$$

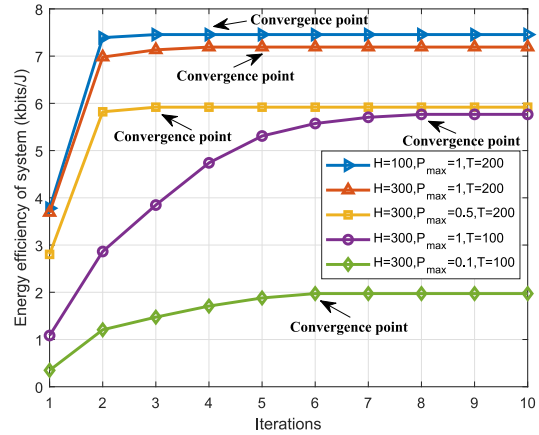

Fig. 3. System EE versus the iteration number of algorithm.

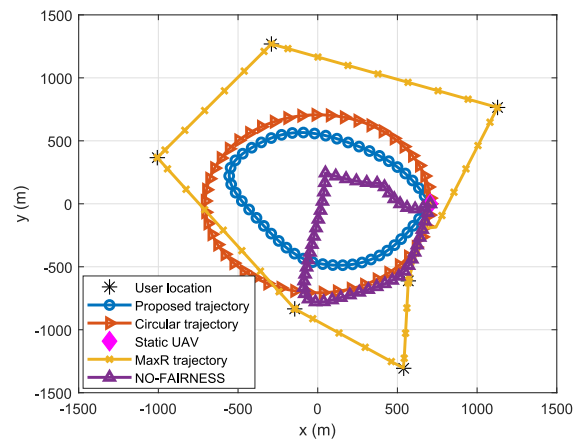

Fig. 4. UAV trajectories of different schemes.

We note that RSD ≥ 0 and $\overline{R_k} \triangleq (\sum_{k=1}^{K}\sum_{n=1}^{N} R_k[n])/K$ is the mean of all users' rate. The lower the value of RSD, the fairer it is between users.

The convergence performance of the proposed J-APQ-maxEE scheme based on different system parameters is investigated, and the result is shown in Fig. 3. It is found that the proposed scheme converges within 3-8 iterations for different parameter settings. It can be found that changes in the flight altitude H and maximum transmit power $P_{\max}$ have little effect on convergence. It can converge in 3-5 iterations although we adjust the altitude and maximum transmit power. But the change in T, UAV flight period, has an obvious effect on the convergence. When T becomes small, it slows down the convergence speed and makes it converge in 6-8 iterations. Nonetheless, it can be seen that the convergence of the proposed algorithm is relatively stable.

The UAV trajectories under different schemes are simulated and the results are illustrated in Fig. 4. The legend "Proposed trajectory" denotes the UAV trajectory obtained by the proposed J-APQ-maxEE, "Circular trajectory" denotes the designed initialization trajectory, "Static UAV" represents the static hovering point, "MaxR trajectory" denotes the trajectory obtained by J-APQ-maxR. We find that under our proposed scheme, the UAV does not pass through all users as it does in the MaxR trajectory scheme. Instead, the proposed trajectory

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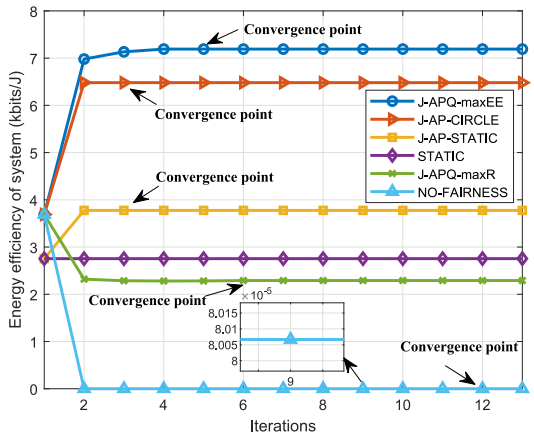

Fig. 5. System EE of different schemes.

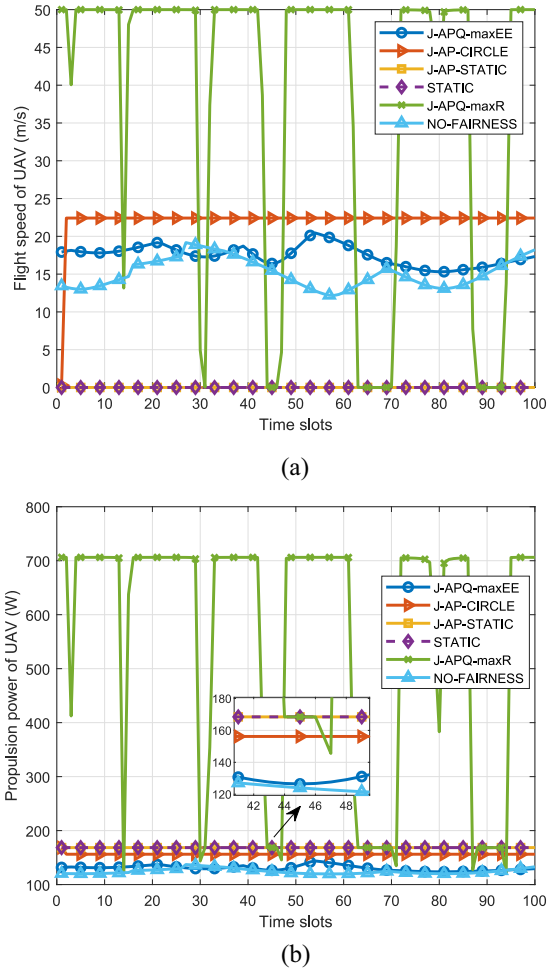

Fig. 6. UAV speed and corresponding UAV propulsion power consumption of different schemes versus time slots. (a) UAV speed. (b) UAV propulsion power consumption.

just roughly coincides with the distribution of users, and even has the shortest flight distance. The reason is that when the time slot length is fixed, the longer flight distance generally corresponds to the higher flight speed, and too high-flight speed results in significant propulsion power consumption as shown in Fig. 2, thus reducing the system EE. We can find that the trajectory of NO-FAIRNESS is mainly concentrated around users 2 and 6, with occasional small proximity to other users. This is because it does not take user fairness into account, in order to increase the sum rate of the users and reduce the UAV's propulsion power consumption, it flies between the closer users at a suitable speed to obtain better channel conditions. The small proximity to the other users also satisfies the basic QoS requirements of them.

To verify the analysis above, the comparisons of the system EE in different schemes are illustrated in Fig. 5. It can be seen that the proposed J-APQ-maxEE achieves the highest system EE (about 0.71 kbits/J, 3.42 kbits/J, 4.43 kbits/J and 4.90 kbits/J higher than the J-AP-CIRCLE, J-AP-STATIC, STATIC and J-APQ-maxR, respectively). The comparison between J-APQ-maxEE and J-APQ-maxR indicates that although the UAV flying over each user can obtain better communication channel conditions, it significantly reduces the system EE as the UAV generates a large amount of propulsion power to fly close to users. This confirms the analysis above. In addition, we find that J-AP-CIRCLE yields suboptimal system EE. This also proves the effectiveness of our designed initial circular trajectory. Furthermore, the comparison results of J-APQ-maxEE with J-AP-CIRCLE and J-AP-STATIC demonstrate the necessity of optimizing UAV trajectory in enhancing EE. The comparison between J-AP-STATIC and STATIC also manifests the effect of resource allocation optimization in improving EE. In particular, the NO-FAIRNESS has the lowest system EE. This is due to the fact that this scheme considers increasing the total user rate rather than balancing between users, and for users that are far away, only their basic rate QoS requirements can be satisfied. So it also produces the user with the lowest rate, based on which the system EE value obtained will be very small. This also proves the important role played by user fairness in system EE.

Further, in order to have a more intuitive understanding of the UAV's flight behavior and propulsion power consumption under different schemes, we give the variation of the UAV flight speed and its propulsion power under different time slots, as shown in Fig. 6. It is seen in Fig. 6(a) that the UAV speeds of the proposed J-APQ-maxEE and NO-FAIRNESS, considering optimizing UAV trajectory to reduce propulsion power consumption, are in the range of 12.2–20.5 m/s, which is around the optimal speed V_{me} in Fig. 2 that corresponds to the minimum propulsion power consumption. Thus, it is not surprising that the UAV under these two schemes consumes the lowest propulsion power in Fig. 6(b). The reason why NO-FAIRNESS consumes slightly less propulsive power than J-APQ-maxEE in many of the time slots is that it does not need to fly close to each user in turn to ensure user fairness, and has fewer restrictions on UAV trajectory. In contrast, Fig. 6(a) shows that the UAV under the J-APQ-maxR often flies at a high speed of 50 m/s to reach and serve each user. As a result, we can easily find from Fig. 6(b) that its high-flight speed leads to as much as 706 W of the propulsion power consumption, which will undoubtedly greatly reduce the system EE. Moreover, by comparing the proposed scheme

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TABLE III
COMPARISON OF JFI VALUES FOR USER FAIRNESS

Simulated Schemes	Value of JFI
J-APQ-maxEE	1
J-AP-CIRCLE	1
J-AP-STATIC	1
STATIC	0.932
J-APQ-maxR	1
NO-FAIRNESS	0.299

TABLE IV
COMPARISON OF RSD VALUES FOR USER FAIRNESS

Simulated schemes	Value of RSD
J-APQ-maxEE	4.0732×10^{-8}
J-AP-CIRCLE	2.2272×10^{-7}
J-AP-STATIC	5.3907×10^{-7}
STATIC	45.0387
J-APQ-maxR	2.3031×10^{-7}
NO-FAIRNESS	417.3286

with J-AP-CIRCLE, J-AP-STATIC and STATIC, we can find that either too high or too low speed of the UAV will result in relatively high-propulsion power consumption, and the proposed scheme can choose the appropriate UAV speed through trajectory optimization to significantly reduce the UAV propulsion power consumption.

In order to verify the fairness of the users, we present the JFI values for the different schemes in Table III. We find that all the schemes, except for STATIC and NO-FAIRNESS, achieve nice fairness, i.e., $JFI = 1$. This is because these two schemes do not consider fairness among users when performing optimization. In addition, we also present the values of RSD in Table IV. As we can see, the RSD value obtained by the proposed J-APQ-maxEE is the smallest. Particularly, the fairness performance of J-APQ-maxEE is better than that of J-AP-CIRCLE, J-AP-STATIC and J-APQ-maxR which also take user fairness into account. Moreover, the RSD values of STATIC and NO-FAIRNESS are on the large side and have poor user fairness. Combined with Fig. 5, we find that the proposed scheme not only can significantly improve the system EE but also achieves the best user fairness.

We note that the EE value produced by the NO-FAIRNESS scheme differs by five orders of magnitude from the EE values of the other schemes and almost close to 0, which can be found from Fig. 5. Therefore, the changes in the subsequent system parameters have a very light effect on its EE values and the numerical changes are very small, so we have ignored this scheme in the latter performance comparisons.

Fig. 7 illustrates the relationship between the system EE and the maximum transmit power of UAV. As the available power resources increase, the EE of all schemes exhibits a continuous improvement. The reason is that when more power resources are available, the UAV can take full advantage of power allocation to enhance the system EE. Consistently, the proposed scheme achieves the highest system EE compared to the other four schemes. We find that the proposed

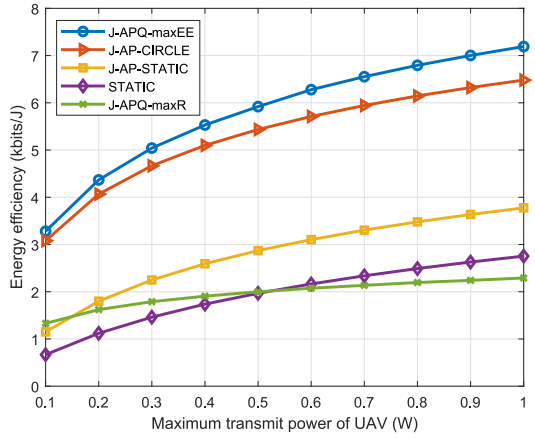

Fig. 7. System EE versus maximum transmit power of UAV.

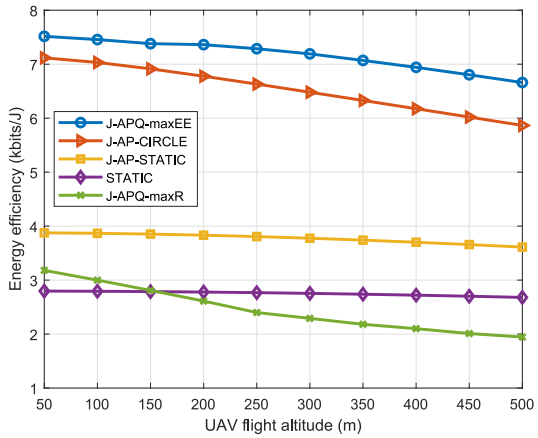

Fig. 8. System EE versus UAV flight altitude.

J-APQ-maxEE can improve the EE by 6.5%–11.0% through trajectory optimization over J-AP-CIRCLE which only uses the designed initialization circular trajectory. In addition, it improves system EE by 90.5%–185.1% and 161.1%–391.4% compared to J-AP-STATIC and STATIC, respectively, through taking advantage of the mobility of UAV. We note that J-APQ-maxEE which involves UAV propulsion power consumption in the optimization can improve system EE by 146.7%-213.9% over J-APQ-maxR.

Fig. 8 demonstrates the impact of the UAV's flight altitude on the system EE. It is found that as the altitude increases, the achieved EE of all schemes decreases. The reason lies in the fact that as UAV goes up in altitude, the path loss is more serious, which will reduce system EE. Notably, the proposed J-APQ-maxEE consistently demonstrates superior performance compared to other schemes. We find that the proposed scheme improves the system EE by 5.6%-13.6%, 84.4%-93.9%, 148.6%-168.7%, and 136.3%-242.4% compared to J-AP-CIRCLE, J-AP-STATIC, STATIC, and J-APQ-maxR, respectively.

The system EE versus UAV flight period is illustrated in Fig. 9. We can find that the performance trends of J-AP-STATIC and STATIC do not change with different flight period. This is because the UAV in both schemes remain stationary during all time slots and is not affected by flight

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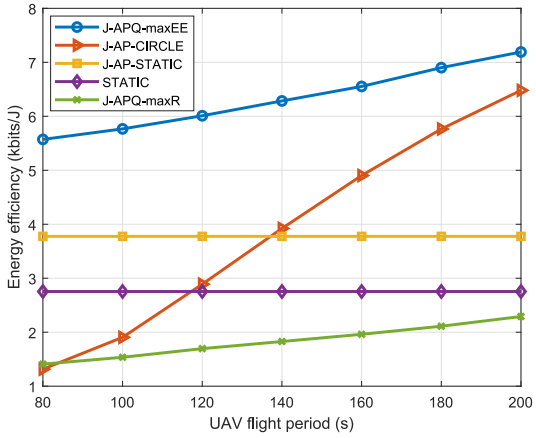

Fig. 9. System EE versus UAV flight period.

period. In contrast, schemes of J-APQ-maxEE, J-AP-CIRCLE, and J-APQ-maxR show continuous improvement in system EE when the flight periods increases. This improvement is due to the fact that longer flight period allows the UAV to spend more time flying close to each user, which improves the user service experience and the overall system EE. It can be found that the proposed J-APQ-maxEE improves the system EE by 11.0%-323.6%, 47.6%-90.5%, 102.3%-161.1%, and 213.9%-295.4% in comparison with J-AP-CIRCLE, J-AP-STATIC, STATIC, and J-APQ-maxR, respectively.

V. Conclusion

This work devoted itself to investigating the energy-efficient problem of the UAV assisted wireless communication network in which a single rotary-wing UAV provides downlink services to multiple users. In doing so, an EE maximization problem is formulated by jointly designing the power allocation, bandwidth assignment and trajectory optimization. In particular, the fairness guarantee of users, the propulsion and communication power consumption of the rotary-wing UAV are all taken into consideration. To address the formulated nonconvex fractional program problem effectively, we initially solve its nonsmoothness. Subsequently, we devise a heuristic algorithm based on the BCD method, which iteratively solves the joint power and bandwidth allocation subproblem and the trajectory optimization subproblem. Additionally, we propose a low-complexity iterative algorithm leveraging Dinkelbach's algorithm and the Lagrange duality technology to attain the optimal solution for the former subproblem. We employ SCA technology and Dinkelbach's algorithm to obtain a suboptimal solution of the latter subproblem. Extensive simulation results demonstrate that the proposed algorithm achieves superior system EE gains while ensuring user fairness, outperforming other benchmark schemes.

Appendix

Proof of Proposition 1

We can find that the objective (16a) is a quasi-convex function because it includes linear numerator and linear denominator. Besides, it is observed that (16b)–(16e) are all affine constraints. Thus, we only need to prove that (16f) is convex. To do that, we observe that the left term $\zeta_k[n]B\log_2(1+[(p_k[n]H_k[n])/(\zeta_k[n])])$ of inequality (16f) is a function containing the variables $p_k[n]$ and $\zeta_k[n]$. For simplicity, we denote this function as $h(x,y) = ax\log_2(1+[by/x])$, where $a \triangleq B$, $b \triangleq H_k[n]$, $x \triangleq \zeta_k[n]$, and $y \triangleq p_k[n]$. The Hessian matrix of function $h(x,y)$ is defined as

$$H = \begin{bmatrix} \frac{\partial^2 h}{\partial x^2} & \frac{\partial^2 h}{\partial x \partial y} \\ \frac{\partial^2 h}{\partial y \partial x} & \frac{\partial^2 h}{\partial y^2} \end{bmatrix}. \tag{51}$$

By calculating the second-order derivative of the function $h(x,y)$, we can obtain that

$$\begin{aligned} H &= \begin{bmatrix} \frac{-ab^2y^2}{x(x+by)^2\log(2)} & \frac{ab^2y}{(x+by)^2\log(2)} \\ \frac{ab^2y}{(x+by)^2\log(2)} & \frac{-ab^2x}{(x+by)^2\log(2)} \end{bmatrix} \\ &= \begin{bmatrix} -y^2 & xy \\ xy & -x^2 \end{bmatrix}. \end{aligned} \tag{52}$$

Next, we find the eigenvalue σ of matrix H from

$$\begin{aligned} |H - \sigma \mathrm{E}| &= \begin{vmatrix} -y^2-\sigma & xy \\ xy & -x^2-\sigma \end{vmatrix} \\ &= \sigma\left(\sigma + (x^2+y^2)\right) = 0. \end{aligned} \tag{53}$$

Then, these two eigenvalues can be derived as

$$\sigma_1 = 0,\ \sigma_2 = -\left(x^2+y^2\right) \le 0. \tag{54}$$

Therefore, $h(x,y)$ exhibits a joint concavity about both x and y, which indicates that $\zeta_k[n]B\log_2(1+(p_k[n]H_k[n])/(\zeta_k[n]))$ demonstrates a joint concavity about $\zeta_k[n]$ and $p_k[n]$. Since the left-hand-side in (16f) is the sum of these concave functions, (16f) is convex. In conclusion, (16) has been proven to be a quasi-convex fractional optimization problem.

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Jing Wang (Student Member, IEEE) received the B.E. degree from the School of Information Science and Engineering, Shandong Normal University, Jinan, China, in 2018. She is currently pursuing the Ph.D. degree with the School of Information Science and Engineering, Shandong University, Qingdao, China.

Her research interests include UAV assisted communications and radio resource management.

Haixia Zhang (Senior Member, IEEE) received the B.E. degree from the Department of Communication and Information Engineering, Guilin University of Electronic Technology, Guilin, China, in 2001, and the M.Eng. and Ph.D. degrees in communication and information systems from the School of Information Science and Engineering, Shandong University, Jinan, China, in 2004 and 2008, respectively.

From 2006 to 2008, she was an Academic Assistant with the Institute for Circuit and Signal Processing, Munich University of Technology, Munich, Germany. From 2016 to 2017, she was a Visiting Professor with the University of Florida, Gainesville, FL, USA. She is currently a Distinguished Professor with Shandong University, Jinan, China. She is actively participating in many professional services. Her research interests include wireless communication and networks, Industrial Internet of Things, wireless resource management, and mobile edge computing.

Dr. Zhang is/was an Editor of the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE WIRELESS COMMUNICATION LETTERS, and *China Communications* and serves/served as a Symposium Chair, a TPC Member, a Sessions Chair, and a Keynote Speaker for many conferences.

Xiaotian Zhou (Member, IEEE) received the B.E. degree in electronic information engineering and the Ph.D. degree in communication and information systems from Shandong University, Jinan, China, in 2007 and 2013, respectively.

He is currently a Full Professor with the School of Control Science and Engineering, Shandong University. His research interests include 5G/B5G wireless communication networks, non-orthogonal multiple access, massive multiple-input–multiple-output, and mobile edge computing.

Prof. Zhou is also an Associate Editor of *IET Communications*.

Wenjie Liu (Graduate Student Member, IEEE) received the B.E. degree from the School of Information Science and Engineering, Shandong University, Jinan, China, in 2017, where he is currently pursuing the Ph.D. degree with the School of Control Science and Engineering.

His research interests include joint caching, computing, and communication resource optimization in MEC networks.

Dongfeng Yuan (Senior Member, IEEE) received the M.S. degree from the Department of Electrical Engineering, Shandong University, Jinan, China, in 1988, and the Ph.D. degree from the Department of Electrical Engineering, Tsinghua University, Beijing, China, in January 2000.

He is currently a Full Professor with the School of Qilu Transportation, Shandong University. From 1993 to 1994, he was with the Electrical and Computer Department, University of Calgary, Calgary, AB, Canada. He was with the Department of Electrical Engineering, University of Erlangen, Erlangen, Germany, from 1998 to 1999; with the Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, MI, USA, from 2001 to 2002; with the Department of Electrical Engineering, Munich University of Technology, Munich, Germany, in 2005; and with the Department of Electrical Engineering, Heriot-Watt University, Edinburgh, U.K., in 2006. His current research interests include intelligent communication systems, mobile-edge computing and cloud computing, and AI and big data processing for communications.

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